

Improving Company Performance by The Correctness of Management Decision through Implementation Dashboard using Power BI Tools (Case Study at Company Y)

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Abstract—Making the right decisions is one of the important things in supporting the performance of a company. Companies need good data analysis in order to produce insightful information that can be used as the basis for making the right decisions. Data needs to be extracted, processed and visualized on analytical panels (dashboard) so that it can be analyzed and become useful information for company management in making decisions. This paper is about dashboard implementation using Power BI (Business Intelligence) in a case study company Y. The goals is to understand how dashboard implementation using Power BI and improve business performance with precision management decisions that can help business enterprises to look back in history and assist with forward planning.

Keywords—microsoft power BI, business intelligence, big data, visualization, dashboard

I. INTRODUCTION

The company is a place for management to operate. In the company, management becomes a subsystem of the company system. Management is a technique or tool that is responsible for estimating, planning, ordering, regulating and controlling effective planning and regulatory operations in a company to achieve the desired goals. And one of the main roles or duties of a manager is in making decisions.

An important aspect of management in a company is business planning and forecasting to have a foresight about the company's business and the resources that should be allocated to drive business performance. This can be achieved by leveraging the data sources that the company stores across multiple platforms where this is often a very difficult and tedious task. Whereas the use of data sources or often called "big data" will produce information that can be used for business analysis and presentation in the future [1].

This paper is about dashboard implementation using Power BI (Business Intelligence) case study in company Y. This study will also show how Power BI (Business Intelligence) as a tool and data source located in various file locations can work together with each other so that become a major equipment reliability reporting system for business analysis and perform real time presentations where this will certainly help in improving management in making the right decisions.

Good data management can make the company take the right decision. Meanwhile, the available data has not been processed so that it is difficult to analyze. One of them is in terms of describing the major equipment reliability in a company. The weakness of the existing system is that in its implementation the management still finds it difficult to monitor the main equipment reliability due to the limited features of the system used.

It takes a database that is useful for grouping interconnected data (big data) in such a way that it can be used quickly and easily. The database is maintained and created by a set of application programs specifically designed to solve a problem, namely by using a system called Sistem Manajemen Basis Data (SMBD). Sistem Manajemen Basis Data is also defined as a computer program that can be used to enter, modify, delete, manipulate, and obtain data or information in a practical and efficient manner [2].

In this case, a Business Intelligence (BI) system is needed to collect, process, and analyze data so that later the results can be estimated by the company and the results can be updated regularly or in real-time [3]. Business Intelligence (BI) is a concept and method of how to improve the quality of business decision making based on data-based systems or it can be said that BI is a data-based decision support system. BI relies heavily on data to be able to perform analysis which is then converted into a new opportunity or breakthrough to develop the company's business. [2]. BI is often equated with business analysts. The basic difference between the two is that business analysts only have the task of analyzing data while BI is obliged to present the results of the analysis. The Power BI application is needed because it can display illustrations so that it can make it easier for those who need to be able to process fast decision making with good quality [4]-[7]

II. METHOD

A. Microsoft Power BI

Microsoft Power BI is a data analytics platform for individual (self-contained), team and organizational services. Microsoft Power BI consists of Power BI Services, Power BI Mobile, and Power BI Desktop. Microsoft Power BI can display information more interactively. Microsoft Power BI can combine, analyze, visualize, and share data. All of these

features can be packaged in a more intuitive and interactive display.

B. Pureshare

Pureshare method is one of the dashboard development methods developed by pureshare vendors to facilitate projects related to measuring and managing the performance of an organization or agency. The pureshare method uses two approaches, namely top-down design and bottom-up implementation. The steps in the pureshare method are planning and design (highlighting the planning and design stage), system and data review (highlighting system and data review), prototype design (highlighting the prototype stage), prototype improvement (highlighting the refinement stage), release, continuous improvement [8].

C. Dashboard Design as a Tool to Improve Company Performance with the Correctness of Management Decisions

This study conducted a qualitative case study to find out the changes that occurred in the business of "Company Y" after the implementation of business intelligence. The main factor of the qualitative case study to be analyzed, is to understand the current situation of "Company Y" by looking at the major daily equipment reports, weekly equipment reports, equipment downtime reports and oil production loss (LPO) reports. "Company Y" has data from different data sources. Data from these various sources will be analyzed using Microsoft Power BI. Data will be extracted using BI tools to be compiled and analyzed using filters that are presented in such a way that adapts to requests from management or users so that they can be used and utilized to support business needs. The benefit and power of using Microsoft Power BI tool is that it can collect data from various sources and it is compiled into 1 integrated data package which can then be filtered as needed very quickly without affecting the original uploaded data. The features found in Microsoft Power BI will help company Y in compiling a summary of the major equipment reliability reports and visual presentations in real time, to help strengthen the analysis of problems that occur in major equipment so that management can make decisions quickly and accurately for business progress (Figure 1).

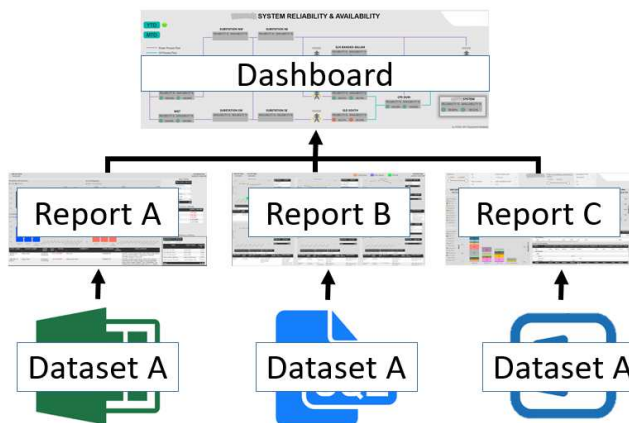


Fig. 1. Power BI dashboard and report summary

III. RESULT AND DISCUSSION

In Power BI, different types of data and data sources can be like Files, Databases, Azure, Online Services and more. Other data sources that can be retrieved are through queries (if

the data sources are other than files) or selecting files from a folder. Data will be loaded in Microsoft Power BI and before creating visuals in Microsoft Power BI. Uploaded data should be analyzed and corrected first to see the relationship between the data. Uploaded data errors can be fixed by using the edit query feature in the data section of Microsoft Power BI. We can perform relationships between data sets from different sources through the manage relationship feature. And if we need conditional columns or unique columns to connect these various data sources, we can take advantage of the column duplication and column append options through the query edit function. After the data is good, neat and integrated, we can visualize the data on a blank screen that has been provided by Microsoft Power BI. The data to be analyzed and corrected can be visualized through different Visuals by utilizing Column Visuals in Microsoft Power BI such as stacked bar charts, stacked column charts, line charts, pie charts and many other visuals available. In Microsoft Power BI visuals that we have created for analysis or called dashboards can be extracted into a report by utilizing the Publish option function so that the visuals that we have created are published in Power BI Services. The dashboard that has been published on the Power BI service in the form of a website can be viewed by anyone and anywhere by providing access to the Power BI Service [8]-[10].

At this stage, several processes are carried out to collect data that will be used in the application of the dashboard using the pureshare method for the case study of company Y, which are as;

- Literature Study

Literature study is used to collect library data and process research materials. This process is carried out to examine more deeply about the dashboard and the pureshare method.

- Data Collection from Company Y Database

In this process, data collection is stored in the Company Y database. The data obtained will later be used to analyze the need to build a dashboard to improve company performance with the correctness of management decision.

A. Analysis Method

The stages contained in the pureshare method [11] are as follows:

1) *Planning and design stage highlights*: This stage is carried out to analyze user needs and can communicate with users to determine what will be displayed on the dashboard.

2) *System and data review highlights*: At this stage, it will be done by identifying data sources and collecting feedback to be developed and adapted to user needs.

3) *Prototype stage highlights*: At this stage it will be designed to build a dashboard by taking a top-down and bottom-up approach simultaneously.

4) *Repair prototype (refinement stage highlights)*: Based on a series of prototypes that have been built, a review will be carried out to get feedback for development according to user needs.

5) *Release*: When the dashboard has passed the testing phase by the user, the dashboard will be implemented and used by Company Y.

6) *Continuous improvement*: This stage is the stage for developing and disseminating dashboards to other areas within an organization if needed

B. Data processing and calculation

In this methodology there are three types of modules as follows Reliability Module, Equipment Down Module and LPO Module. Reliability module contains major equipment Reliability submodule, Equipment Down Module contains running/shut down submodule of major equipment including reasoning and comment based on actual condition and LPO module contains running/shut down of oil well production. Pseudo Code Algorithms:

Pseudo Code Algorithm 1:

```

Start
Get 'Qi'
View 'Qi'
Initialise Qa, Qb, Qc, Qd.
Processing of Qi by means of string operations
Edit the dataset 'Qi' for adding new conditional columns
Qrun, Qava, Qrel, Qgc using edit query function
Create new column Qrun
Qrun = 1 - Qa / 24;
Create new column Qava
Qava = 1 - Qb / 24;
Create new column Qava
Qrel = 1 - Qc / 24;
Create new column Qgc
Qgc = If Qd = "AW1", Qrel * 212 / 24,
      Else If Qd = "AW2". Qrel * 110 / 24,
      Else If Qd = "AW5". Qrel * 135 / 24,
      Else blank
Visualize the variables considered above using a different
visual
Publish the Report to Power BI Services notations
Qi = Dataset of Reliability Module
Qa= Variable of running time
Qb= Variable of planned time
Qc= Variable of unplanned time
Qd= Variable of major equipment code
Qrun= Variable of runtime
Qava= Variable of availability
Qrel= Variable of reliability
Qgc= Variable of LPO compressor

```

Pseudo Code Algorithm 2:

```

Start
Get 'Qii'
View 'Qii'
Initialise Qe, Qf, Qg, Qh
Processing of Qii by means of string operations
Edit the dataset 'Qii' for adding new conditional columns
Qtyp, Qsts, Qcol, Qdff using edit query function
Create new column Qtyp
Qtyp = Qe & "-" & Qf
Create new column Qsts
Qsts = If Qg = 04, CONSTRAIN
      Else If Qg = 11, PM
      Else If Qg = 13, FAIL
      Else If Qg = 543, S_FAIL
      Else Qg
Create new column Qcol
Qcol = If Qsts = CONSTRAIN, #FFFF00
      Else If Qsts = PM, #0000FF
      Else If Qsts = FAIL, #FF0000
      Else If Qsts = S_FAIL, #000000
      Else blank
Create new column Qdff
Qdff = Qh & "- Days
Visualize the variables considered above using a different
visual
Publish the Report to Power Bi Services notations
Qii = Dataset of Equipment Down Module
Qe = Variable of subproduction
Qf = Variable of equipment type
Qg = Variable of description 1
Qh = Variable of days off
Qtyp = Variable of production area
Qsts = Variable of status
Qcol = Variable of coloring
Qdff = Variable of total days off

```

Pseudo Code Algorithm 3:

```

Start
Get 'Qiii'
View 'Qiii'
Initialise Qi, Qj, Qk, Ql, Qm, Qn, Qo, Qp, Qr, Qs
Processing of Qiii by means of string operations
Visualize the variables considered above using a different
visual

```

Publish the Report to Power Bi Services notations

Qiii = Dataset of LPO Module

Qi = Variable of production area

Qj = Variable of asset id

Qk = Variable of asset name

Ql = Variable of event category

Qm = Variable of event reason

Qn = Variable of event system

Qo = Variable of oil loss

Qp = Variable of water loss

Qr = Variable of facility

Qs = Variable of subsurf

In the Pseudo code Algorithm 1, 2, and 3 which are different modules. First the tool program is started and the required data like Qi , Qii, Qiii in Algorithms 1, 2 and 3 are fetched from different data sources via datamart and edit queries. then each dataset is viewed and corrected in certain parts so that the dataset is good, neat and structured and then the data is initialized with several columns such as Qrun, Qava, Qrel, Qgc, Qtyp, Qsts, Qcol, Qdff as shown in the Pseudo-Code 1,2, and 3. In each Pseudo-code a different dataset, contains a different number of columns as well. In some parts, so that each dataset originating from several sources is integrated with each other, conditional columns are needed as connectors [12]-[15]. Conditional columns can be created using the edit query function as shown in pseudocode 1 and 2. Conditional columns can be mathematical operations and other operations. then the integrated dataset is visualized through Microsoft Power BI Visuals [16]. The visualized data or commonly called a dashboard is then published to BI Services through the Publish function. After publishing the dashboard to the Power BI service, the resulting link can be shared with anyone provided that access has been granted and can be shared with individual organizations via the Uniform Resource Locator (URL) or it can be via SharePoint, Websites, etc, as shown in the section below results and discussion [17].

In this section, the results of the dashboard development that have been carried out will be discussed. This research produces a dashboard to improve company performance by the correctness of management decision. The data used comes from the Company Y database. The data is separated into several tables based on the type. Starting from the LPO table, equipment down table, well down table, communication table, reliability and availability table. From each of these tables, a chart can be made which, when put together, will form a dashboard.

C. Dashboard Design of Company Y

In this section, we will explain the steps involved in designing a dashboard.

1) *Planning and design stage highlights:* The first stage is planning and design to obtain a basic picture of the needs of an agency. At this stage there are several activities to analyze dashboard needs including:

a) *Identifying Goals:* The purpose of this dashboard design is to help present important information contained in

the company Y database in an attractive, effective and efficient visual form so that information can be easily understood.

b) *Determining Key Performance Indicators (KPI):* The KPIs that will be used in the dashboard in this research are the percentage of reliability and availability of equipment, the amount of lost production oil, equipment down reason record and well down reason record.

2) *System and data review highlights:* This stage is carried out to identify the data sources that will be used. The data must be in accordance with the needs in the dashboard development. The data used is obtained from the database of company Y.

3) *The dashboard prototype design is done using Microsoft Power BI:* This process is carried out to process data in order to produce information as needed.

a) *Tidy Up Data Sources:* Source of data received from database of company Y using oracle database, SQL Server etc. When viewed, the data consists of many tables that have not been arranged. So that the data must be tidied up first to make it suitable and have value or meaning so that it can be visualized on the dashboard.

b) *Data processing:* This process is carried out to process data in order to produce information as needed.

- Create New Script (query) and Copy Data

Creating new scripts and copying data aims to prevent the original data from being mixed up with the processing so that if something goes wrong, the original data is not lost or messy. Can be seen at Figure 2 below.

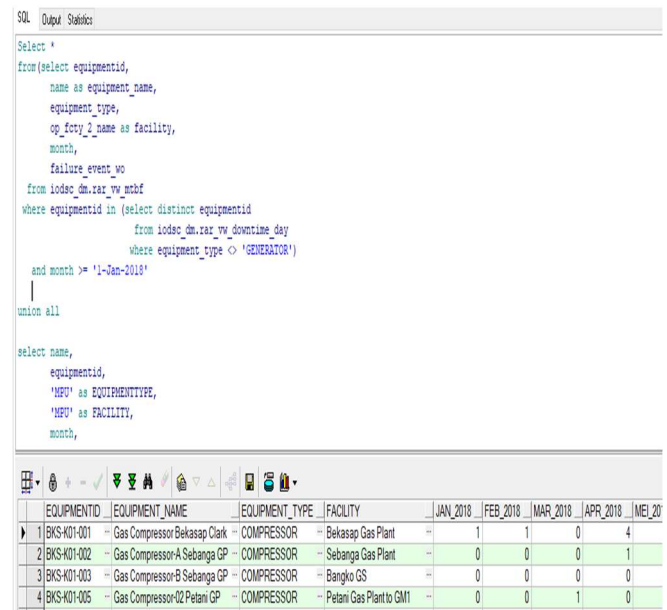
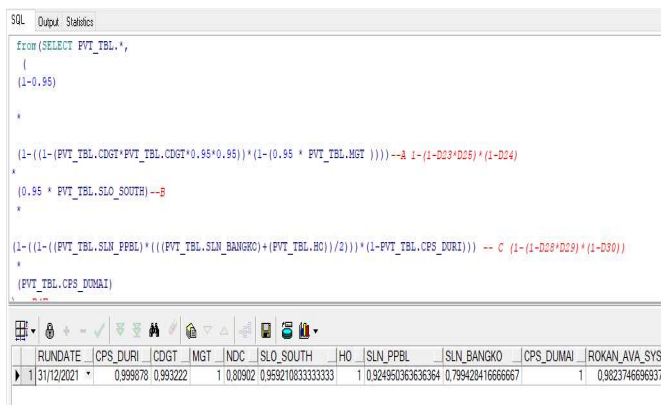


Fig. 2. Creating new script (query)

- Calculation Process

After the data is copied into a new script (query), the calculation process is carried out according to what was generated. Can be seen at Figure 3.



- Importing Data In Microsoft Power BI

When all the required calculations are done, the next step is to import the excel data into Microsoft Power BI. Can be seen at Figure 4.

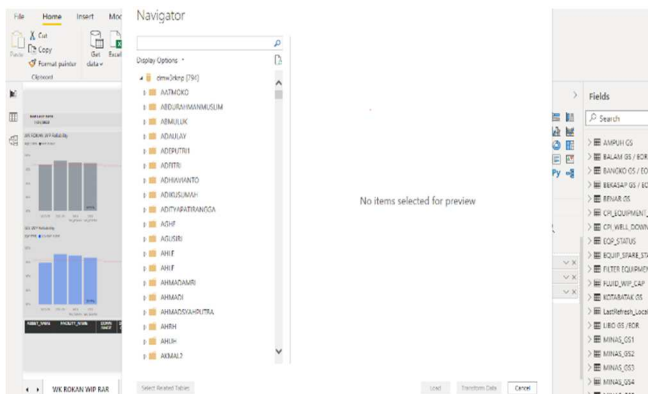


Fig. 4. Importing data in microsoft power BI

The design is made based on the results of the needs analysis that has been done [18], [19]. After publishing the dashboard to the power BI services it appears as shown in below figure and we can have some editing operations if any filters or visuals are not properly accessible then they can be rectified here before publishing on to the web [20]. following are the results of dashboards of discussed modules of Company Y as in Figure 5, Figure 6 and Figure 7.

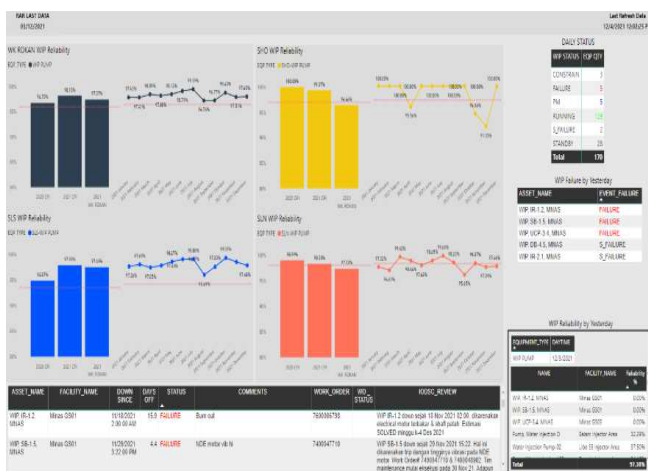


Fig. 5. Major equipment pump summary dashboard



Fig. 6. Equipment down and LPO dashboard

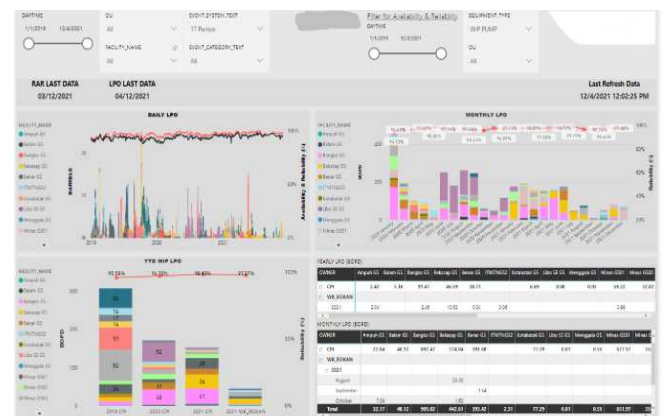


Fig. 7. LPO summary Dashboard

IV. CONCLUSION

From these results, it can be said that the presentation of data and information regarding the Major Equipment Reliability data in company Y can use Power BI. The data that has been obtained from several sources can be used as a reference to develop of Major Equipment Reliability Dashboard at Company Y. The characteristics of the results of this study whose contributions consist of 3 dashboards, Major Equipment Pump Summary Dashboard, Equipment Down and LPO Dashboard and LPO Summary Dashboard.

The data collected on each dashboard contains information that can be analyzed, so that information can be obtained that can be used by management as a reference in making the right decisions. Dashboards that have been created can be viewed in Power BI Desktop or Power BI Services so management can view information about Major Equipment Reliability that can help business enterprises look back on history and help plan ahead.

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